**FORMULAE LIST**

**Circle:**

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle centre $(a, b)$ and radius $r$.

**Scalar Product:** $\boldsymbol{a}.\boldsymbol{b} = |\boldsymbol{a}|\,|\boldsymbol{b}| \cos\theta$, where $\theta$ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$

or $\boldsymbol{a}.\boldsymbol{b} = a_1b_1 + a_2b_2 + a_3b_3$ where $\boldsymbol{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

**Trigonometric formulae:**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$
$$\sin 2A = 2\sin A \cos A$$
$$\cos 2A = \cos^2 A - \sin^2 A$$
$$= 2\cos^2 A - 1$$
$$= 1 - 2\sin^2 A$$

**Table of standard derivatives:**

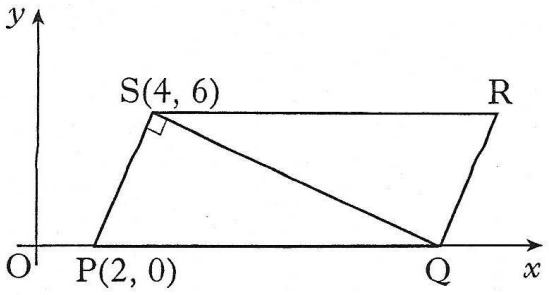
| $f(x)$ | $f'(x)$ |
|---|---|
| $\sin ax$ | $a\cos ax$ |
| $\cos ax$ | $-a\sin ax$ |

**Table of standard integrals:**

| $f(x)$ | $\int f(x)\,dx$ |
|---|---|
| $\sin ax$ | $-\frac{1}{a}\cos ax + C$ |
| $\cos ax$ | $\frac{1}{a}\sin ax + C$ |

**1.** PQRS is a parallelogram. P is the point (2, 0), S is (4, 6) and Q lies on the $x$-axis, as shown.

The diagonal QS is perpendicular to the side PS.



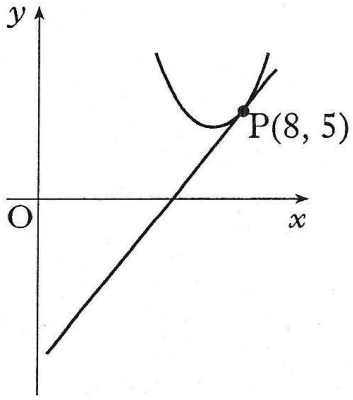

($a$) Show that the equation of QS is $x + 3y = 22$. **4**

($b$) Hence find the coordinates of Q and R. **2**

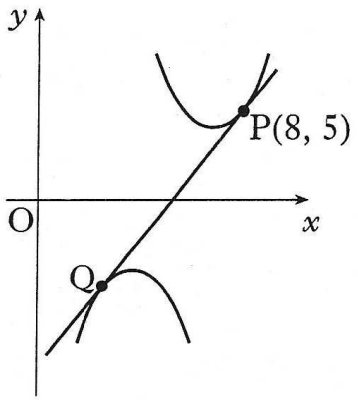
**2.** Find the value of $k$ such that the equation $kx^2 + kx + 6 = 0$, $k \neq 0$, has equal roots. **4**

**3.** The parabola with equation $y = x^2 - 14x + 53$ has a tangent at the point P(8, 5).

($a$) Find the equation of this tangent. **4**




($b$) Show that the tangent found in ($a$) is also a tangent to the parabola with equation $y = -x^2 + 10x - 27$ and find the coordinates of the point of contact Q. **5**




**4.** The circles with equations $(x - 3)^2 + (y - 4)^2 = 25$ and $x^2 + y^2 - kx - 8y - 2k = 0$ have the same centre.

Determine the radius of the larger circle. **5**

**5.** The curve $y = f(x)$ is such that $\frac{dy}{dx} = 4x - 6x^2$. The curve passes through the point $(-1, 9)$. Express $y$ in terms of $x$. **4**

**6.** P is the point $(-1, 2, -1)$ and Q is $(3, 2, -4)$.

(*a*) Write down $\overrightarrow{PQ}$ in component form. **1**

(*b*) Calculate the length of $\overrightarrow{PQ}$. **1**

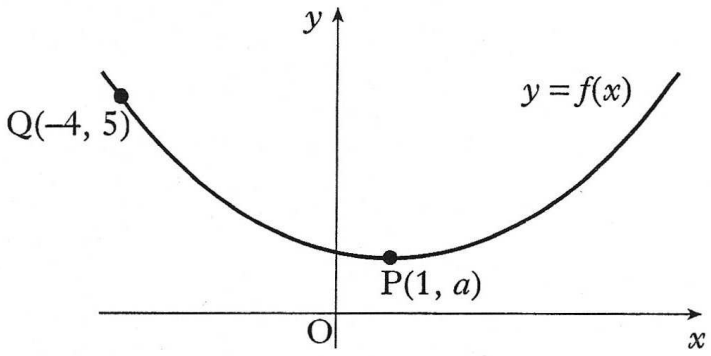
(*c*) Find the components of a unit vector which is parallel to $\overrightarrow{PQ}$. **1**

**7.** The diagram shows the graph of a function $y = f(x)$.

Copy the diagram and on it sketch the graphs of:

(*a*) $y = f(x - 4)$; **2**

(*b*) $y = 2 + f(x - 4)$. **2**



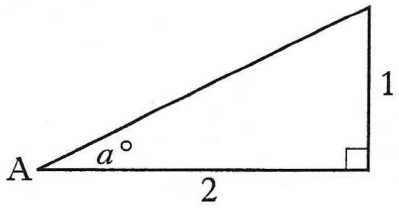

**8.** The diagram shows a right-angled triangle with height 1 unit, base 2 units and an angle of $a°$ at A.

(*a*) Find the exact values of:

(i) $\sin a°$;

(ii) $\sin 2a°$. **4**

(*b*) By expressing $\sin 3a°$ as $\sin(2a + a)°$, find the exact value of $\sin 3a°$. **4**




**9.** If $y = \frac{1}{x^3} - \cos 2x$, $x \neq 0$, find $\frac{dy}{dx}$. **4**

**10.** A curve has equation $y = 7\sin x - 24\cos x$.

(*a*) Express $7\sin x - 24\cos x$ in the form $k\sin(x - a)$ where $k > 0$ and $0 \leq a \leq \frac{\pi}{2}$ **4**

(*b*) Hence find, in the interval $0 \leq x \leq \pi$, the $x$-coordinate of the point on the curve where the gradient is 1. **3**

**11.** *It is claimed that a wheel is made from wood which is over 1000 years old.*

To test this claim, carbon dating is used.

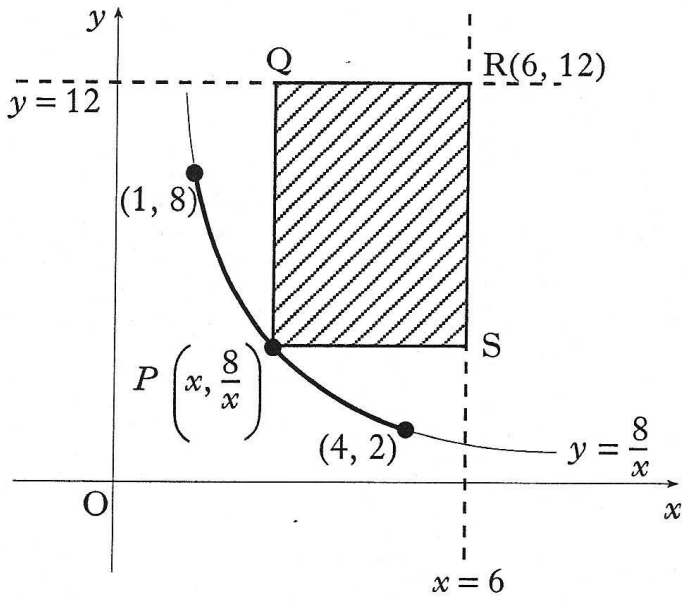
The formula $A(t) = A_0 e^{-0{\cdot}000124t}$ is used to determine the age of the wood, where $A_0$ is the amount of carbon in any living tree, $A(t)$ is the amount of carbon in the wood being dated and $t$ is the age of the wood in years.

For the wheel it was found that $A(t)$ was 88% of the amount of carbon in a living tree.

Is the claim true? **5**

**12.** PQRS is a rectangle formed according to the following conditions:

- it is bounded by the lines $x = 6$ and $y = 12$
- P lies on the curve with equation $y = \frac{8}{x}$ between (1, 8) and (4, 2)
- R is the point (6, 12).




(*a*) (i) Express the lengths of PS and RS in terms of $x$, the $x$-coordinate of P.

(ii) Hence show that the area, A square units, of PQRS is given by

$$A = 80 - 12x - \frac{48}{x}.$$ **3**

(*b*) Find the greatest and least possible values of A and the corresponding values of $x$ for which they occur. **8**